MODULE 07 · DOMAIN 2 - OUTPUT EVALUATION AND VALIDATION

Hallucinations, Inconsistencies and Bias

Three ways a fluent output can still be wrong: how each one happens, how to spot it, and what Anthropic recommends to reduce it.

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A hallucination is not a lie. It is a confident guess that happens to be wrong.

Why hallucinations happen

- **Prediction, not lookup.** Without sources or search, Claude generates what is plausible.
- **Gaps get filled.** Ask for a figure it does not have, and a plausible figure may appear.
- **Tone is constant.** A wrong answer sounds exactly as sure as a right one.
- **Knowledge has a cutoff.** Recent events and your own organisation's facts are invisible unless supplied.

> ON THE EXAM
>
> Anthropic: even the most advanced models can produce text that is factually incorrect or inconsistent with the given context.

Warning signs of a hallucination

WARNING SIGN	EXAMPLE
An unnamed authority	"The recognised industry standard says..."
A suspiciously precise figure	"68.4% of facilities managers..." with no source
A citation you cannot find	A report title, quote or link that does not exist
Facts about you that you never supplied	Your client's contract value, your staff numbers
Recent events, stated confidently	Something after the knowledge cutoff
Evidence that is suspiciously convenient	Exactly the statistic the argument needed

ON THE EXAM

Any one of these means: treat the claim as unverified until you have checked it.

Three different defects

DEFECT	WHAT IT LOOKS LIKE	HOW YOU CATCH IT
Hallucination	A false claim, stated confidently	Check it against a source
Inconsistency	The output contradicts itself	Compare its parts
Bias	Unfair to a group or view	Ask who is favoured or missing

THE TIEBREAKER

A hallucination is checked against the world. An inconsistency is checked against the output itself.

The unnamed standard

- **The weekly report says** Halden beats "the recognised industry standard" for response times.
- **Nobody can find the standard.** It names no body, no document, no figure.
- **The right response:** treat it as unverified. Source it from an authoritative reference, or remove it.
- **Not acceptable:** keeping it because it sounds right, or softening it to "widely recognised".

> ON THE EXAM
>
> An authority without a name is a warning sign, not a source.

Inconsistencies: when the output disagrees with itself

- **The conclusion contradicts the evidence** in the body. The most serious structural defect.
- **Numbers that do not add up:** totals that are not the sum of their parts.
- **Counts that change:** 12 sites in one section, 14 in another.
- **Names and dates that drift:** one client, two spellings; a deadline that moves.

> ON THE EXAM
>
> If the conclusion and the evidence disagree, the output cannot be trusted until it is fixed.

Bias: who is favoured, and who is missing?

- **Anthropic's definition:** systematic patterns that unfairly favour or disadvantage certain groups or perspectives.
- **Generalisations about groups:** "older workers struggle with new apps".
- **Loaded requirements:** "young and energetic" in a job advert.
- **One-sided framing:** only management's view of a change that affects field staff.
- **Missing perspectives:** nobody asked the people most affected.

> ON THE EXAM
>
> Bias in outputs often mirrors bias in the data Claude learned from. Review for it on purpose.

Generalisation, or observation?

A GENERALISATION

What a group is "like"

- "Retail clients are price-obsessed"
- Applies a trait to every member
- Unfair to individuals
- Cannot be checked

AN OBSERVATION

What actually happened

- "12 of 40 retail clients renegotiated on price"
- Describes measured behaviour
- Fair to individuals
- Checked against data

THE TIEBREAKER

Replace claims about what a group is like with what was observed, then check it against data.

Anthropic's ways to reduce hallucinations

TECHNIQUE	WHAT TO ASK FOR
Permission to not know	"If the documents do not say, tell me you do not know."
Quotes first	For long documents: extract word-for-word quotes, then answer
Cite or retract	Support each claim with a quote; drop any claim without one
Stay in the documents	Use only the material provided, not general knowledge
Compare several runs	Ask more than once; disagreement flags a weak claim
Read the reasoning	Review Claude's thinking summary when an answer looks wrong

ON THE EXAM

Allowing "I don't know" alone can drastically reduce false information, says Anthropic.

These techniques reduce hallucinations.
They do not eliminate them.

When you suspect a hallucination

01
Pinpoint
The specific claim that looks unsupported.

02
Check
Against the source or a trusted reference.

03
Act
Correct it, source it, or remove it.

04
Find the cause
Missing material? Unclear scope?

05
Fix upstream
Supply sources, allow "I don't know".

ON THE EXAM

First, pinpoint the exact claim. Discarding the whole output is rarely the right first move.

Claude drafted a client update for a Halden account manager, using the visit log, the rota email and a fault ticket she attached.

Before sending, she has time to check only one statement properly.

Which statement most needs checking before the update is sent?

A "Industry research shows that over two thirds of facilities managers now expect weekly reporting."

B "Your site received four cleaning visits last week, as the attached log shows."

C "The new rota begins on the date you approved in your email."

D "We fixed the boiler fault you reported on Monday."

◆ PAUSE AND DECIDE

Which statement most needs checking before the update is sent?

A "Industry research shows that over two thirds of facilities managers now expect weekly reporting."

Unnamed research, a precise figure, and nothing attached supports it.

B "Your site received four cleaning visits last week, as the attached log shows."

Traceable to the attached log.

C "The new rota begins on the date you approved in your email."

Traceable to the rota email.

D "We fixed the boiler fault you reported on Monday."

Traceable to the fault ticket.

THE SIGNAL

Three statements trace to attached material. One cites research nobody supplied.

> Claude drafted a supplier review for Halden's procurement team. The body shows the window-cleaning supplier missed its service targets in every quarter.
>
> The draft has several other faults too.

Which defect most seriously undermines the review?

A Two section headings use different styles

B The conclusion recommends renewing the supplier's contract

C One point is made twice in different paragraphs

D A summary table has no title

◆ PAUSE AND DECIDE

Which defect most seriously undermines the review?

A Two section headings use different styles

Cosmetic. Easy to fix, harmless to meaning.

B The conclusion recommends renewing the supplier's contract

Contradicts the evidence in the body. The reader cannot trust either.

C One point is made twice in different paragraphs

Untidy, but does not change the meaning.

D A summary table has no title

A presentation fault, not a meaning fault.

THE SIGNAL

Missed targets every quarter, yet the conclusion says renew: the conclusion contradicts the evidence.

A Halden recruiter asks Claude for a job advert for field cleaners. The draft describes the ideal candidate as "young and energetic" and "a native English speaker".

The role involves physical work, and safety briefings are given in English.

Which two actions are most appropriate? Choose two.

A Publish the advert, because Claude writes neutrally by default

B Ask Claude to make the tone warmer and more welcoming

C Replace the age and native-speaker wording with the role's real requirements

D Add more descriptive adjectives about the ideal candidate

E Check the revised advert against Halden's equal opportunities guidance before publishing

◆ PAUSE AND DECIDE

Which two actions are most appropriate? Choose two.

A Publish the advert, because Claude writes neutrally by default

The draft plainly is not neutral. Outputs can carry bias.

B Ask Claude to make the tone warmer and more welcoming

A warmer tone keeps the same unfair requirements.

C Replace the age and native-speaker wording with the role's real requirements

Describes the job: physical tasks, and enough English for safety briefings.

D Add more descriptive adjectives about the ideal candidate

More description of a person, not the job, invites more bias.

E Check the revised advert against Halden's equal opportunities guidance before publishing

Verifies the fix against the organisation's standard.

THE SIGNAL

Requirements describing a person rather than the job: fix the wording, then check it.

> Halden HR uses Claude to answer managers' questions from the attached staff handbook.
>
> Twice this month, an answer included a rule that does not appear anywhere in the handbook.

Which prompt change best reduces this?

A Ask Claude to be more accurate in its answers

B Switch to the most capable model available

C Ask for shorter answers, so there is less room for error

D Answer only from the handbook, quote the section relied on, and say so when the handbook is silent

◆ PAUSE AND DECIDE

Which prompt change best reduces this?

A Ask Claude to be more accurate in its answers

An exhortation. It gives Claude no new way to be accurate.

B Switch to the most capable model available

Any model may fill gaps when allowed to.

C Ask for shorter answers, so there is less room for error

A short invented rule is still invented.

D Answer only from the handbook, quote the section relied on, and say so when the handbook is silent

Three of Anthropic's techniques: restrict, quote, allow "I don't know".

THE SIGNAL

Invented rules appear where the handbook is silent: restrict, quote, and allow "I don't know".

A Halden analyst is reviewing a Claude-drafted board paper on new workplace cleaning regulations. It cites a named government report, with a title and a year.

She searches for the report and cannot find it anywhere.

What should she do?

A Treat the citation as unverified: check the official source, and remove or replace the claim if it cannot be confirmed

B Keep it, because Claude usually gets citations right

C Ask Claude whether the report exists, and keep it if Claude says yes

D Keep the claim but change the wording to "according to some sources"

◆ PAUSE AND DECIDE

What should she do?

A Treat the citation as unverified: check the official source, and remove or replace the claim if it cannot be confirmed

Validates against the authoritative source before the board relies on it.

B Keep it, because Claude usually gets citations right

"Usually" is not good enough for a board paper.

C Ask Claude whether the report exists, and keep it if Claude says yes

A model confirming its own citation is not verification.

D Keep the claim but change the wording to "according to some sources"

Hides the problem behind vaguer wording.

THE SIGNAL

A confident, specific citation that cannot be found: verify at the authoritative source, or remove.

> A Claude-drafted account plan for Halden's retail clients says: "Retail clients are price-obsessed and will leave at the first cheaper offer."
>
> Halden holds renewal records for all forty of its retail clients.

What is the best response?

A Keep it, because it matches what several account managers believe

B Soften it to "many retail clients are price-sensitive"

C Delete the whole account plan and start again

D Replace it with observed behaviour from the renewal records, and check the figures

◆ PAUSE AND DECIDE

What is the best response?

A Keep it, because it matches what several account managers believe

Shared belief is not evidence, and it misjudges individual clients.

B Soften it to "many retail clients are price-sensitive"

Still a generalisation, still unchecked.

C Delete the whole account plan and start again

Overreacts. One sentence needs fixing, not the plan.

D Replace it with observed behaviour from the renewal records, and check the figures

Turns a generalisation into a checked observation.

THE SIGNAL

Real data exists: replace what a group is "like" with what the records show.

In five lines

- **Hallucination:** a confident guess that is wrong. Tone tells you nothing.
- **Warning signs:** unnamed authorities, precise unsourced figures, citations you cannot find.
- **Inconsistency:** the output contradicts itself. Worst case, the conclusion contradicts the evidence.
- **Bias:** replace generalisations with observations, and describe the job, not the person.
- **Reduce, then verify:** allow "I don't know", quote, cite or retract. Nothing eliminates errors.

COMING UP

Next: fact-checking and validation techniques, step by step.